Baptiste AUDUGÉ

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Engineering Student at CentraleSupélec | Applied AI, client-facing

Seeking a 6-month internship beginning in February/March 2027 in Asia and USA

Engineering student with a math & CS background, currently building **physics-informed deep learning models for uranium mining** at Orano. Experienced in turning client problems into shipped software, from freelance interactive art installations to industrial AI.

EXPERIENCE

Orano | *Global nuclear fuel cycle company* — Paris, France
Data Scientist Intern – Deep Learning & Generative AI — Jul 2026 – Jan 2027

- Asked to fix a **production AI model** planning well cleaning across uranium mines; traced its errors to an overlooked factor: **interactions between neighboring wells**.
- Designed a new **physics-informed model**: studied the geology literature and **derived its governing equations**.
- Built the data extraction (SQL, PI Web API) from the mine's monitoring systems; found **~30% of legacy data silently imputed**.
- Implementing the model in PyTorch: **Transformer with spatio-temporal cross-attention** coupled to a **Neural ODE**, trained on 10,000+ well time series spanning up to 20 years.

QSTNMRK | *Art Collective (Freelance)* — Paris, France
Lead Developer — Jan 2024 – Present

- *The Undrinkable Can* (2024): brought in to help on the campaign website; **proposed and delivered a full rebuild**, which led to becoming the collective's lead developer.
- Now own **all technical delivery** across campaigns: code, hosting, payments and transactional email workflows.
- *The Last Dollar* (2026, featured in **designboom**): live-streamed sculpture fed by an internet-connected ATM. Built it end-to-end, including a **24/7 streaming infrastructure** designed for flat cost regardless of audience.

Freelance Full-Stack Developer — Paris, France
Monaliza | *B2B2C financial products distributor, in partnership with* **Generali** — Jun 2022 – Sep 2024

- Worked directly with the founders to build the **real-time simulation engine** (NestJS, React) of the platform brokers integrate to sell the company's products, encoding complex business rules.
- Became the core tool of **investor presentations**, contributing to strategic partnerships.

Lexem | *Wealth Management Firm* — Nov 2020 – Jul 2022

- Built a **tax-impact simulator** and a client-intake platform with real-time product simulation, designed with advisors.
- Key asset in the firm's sales strategy; contributed significantly to its **partial acquisition**.

EDUCATION

CentraleSupélec – Université Paris-Saclay | *Master's-level Engineering Degree (Diplôme d'Ingénieur)* — Paris, France
Coursework: Machine Learning, High-Performance Computing, Statistics, Signal Processing — 2024 – 2028 (expected)

Sorbonne University | *Double Bachelor's Degree in Mathematics & Computer Science* — Paris, France
Coursework: C Programming, Data Structures, Databases, Probability, Algebra — 2021 – 2024

LEADERSHIP & ACTIVITIES

Sono Barco CentraleSupélec | *Student Association* — Paris, France
Treasurer & Event Technician — 2025 – 2026

- Managed **€120k annual revenue**, tax filings (corporate tax, VAT) and the relationship with the accountant.
- Negotiated service contracts with HEC, ESSEC, Mines Paris; led event setups with **€300k+ of equipment**.

ACADEMIC PROJECTS

- **IoT Anomaly Detection (Spark)**: anomaly detection and root-cause pipeline on large-scale IoT data from **Bouygues Telecom**.
- **Graphical Models in Finance**: conditional dependency detection across MSCI World assets using **Graphical Lasso**.

SKILLS & LANGUAGES

Languages: English (C1), French (native), Spanish (B1)
AI & Data Science: Physics-informed deep learning, time-series modeling, statistics, PyTorch, Scikit-learn, Spark
Software & Cloud: Python, TypeScript (NestJS, React), C/C++, Java, SQL, REST APIs, Docker, AWS, Cloudflare, Git, Linux, Claude Code, MCP
Client & Business: Stakeholder requirements gathering, contract negotiation, budgeting & accounting
Interests: Climbing, sound engineering & DJing, bass guitar, aeronautics (French Aviation Certificate, BIA)